Vectors I - HW1 (Introduction to Ratio Theorem)

Questions

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Original practice questions. Total: 52 marks. Give exact answers unless otherwise stated.

Question 1 **[6 marks]**

The points A, B, C and D have coordinates $(1, 2, 0)$, $(4, -2, 0)$, $(-2, 5, 1)$ and $(1, 1, 1)$ respectively.

(a) Find $\overrightarrow{AB}$, $\overrightarrow{CD}$ and the distance AB. [3]

(b) Are $\overrightarrow{AB}$ and $\overrightarrow{CD}$ equal vectors? Are $\overrightarrow{OA}$ and $\overrightarrow{OC}$ equal vectors? Explain the difference between these two comparisons. [2]

(c) A student claims that A, B, C, D are collinear because $\overrightarrow{AB} = \overrightarrow{CD}$. Explain why this is false. [1]

Question 2 **[5 marks]**

The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 4\mathbf{j}$.

(a) Write $\mathbf{a}$ and $\mathbf{b}$ as column vectors and find $\mathbf{a} + 2\mathbf{b}$. [2]

(b) Find $\mathbf{c}$ such that $2\mathbf{a} - 3\mathbf{b} + \mathbf{c} = \mathbf{0}$. [2]

(c) Points P and Q have position vectors $\mathbf{a} + 2\mathbf{b}$ and $\mathbf{c}$ respectively. Find $\overrightarrow{PQ}$. [1]

Question 3 **[5 marks]**

The vector $\mathbf{u} = \begin{pmatrix} t \\ 3 \\ t+2 \end{pmatrix}$ has magnitude $\sqrt{19}$, where t is real.

(a) Find all possible values of t. [3]

(b) Find the corresponding unit vectors in the direction of $\mathbf{u}$. [1]

(c) Are the two possible vectors $\mathbf{u}$ parallel? Justify your answer without calculating any angle. [1]

Question 4 **[6 marks]**

The points A and B have coordinates $(1, -2, 2)$ and $(3, 1, -4)$ respectively.

(a) Find the unit vector in the direction of $\overrightarrow{AB}$. [2]

(b) Find the vector of magnitude 21 in the direction opposite to $\overrightarrow{AB}$. [2]

(c) A point P is such that $\overrightarrow{AP}$ has magnitude 14 and the same direction as $\overrightarrow{AB}$. Find the coordinates of P and state the order of the three points along their common straight path. [2]

Question 5 **[6 marks]**

The vectors $\mathbf{p}=\begin{pmatrix}6\\k\\-3\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}2\\-4\\m\end{pmatrix}$ are parallel, where k and m are real constants.

(a) Find k and m. [3]

(b) State whether $\mathbf{p}$ and $\mathbf{q}$ have the same or opposite directions, and find $|\mathbf{p}|:|\mathbf{q}|$. [1]

(c) Find all vectors of magnitude $\sqrt{21}$ that are parallel to $\mathbf{p}$. [2]

Question 6 **[6 marks]**

The points A, B and C have coordinates $(1, 2, -1)$, $(3, 1, 2)$ and $(7, -1, 8)$ respectively.

(a) Show that A, B and C are collinear, and find $AB:BC$. [3]

(b) The point $R(u, -2, v)$ lies on the same straight path as A, B and C. Find u and v. [3]

Question 7 **[8 marks]**

The points A and B have coordinates $(2, -1, 3)$ and $(-4, 5, 0)$ respectively. A point M divides AB internally in the ratio $AM:MB=2:1$. A point N lies on AB produced beyond B and satisfies $AN:NB=5:2$.

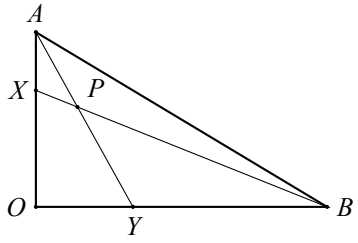
(a) Find the position vector of M. [2]

(b) Find the position vector of N. [3]

(c) State the order of the four points and find $MB:BN$. [3]

Question 8 **[10 marks]**

Relative to the origin O, the points A and B have non-zero, non-parallel position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point X lies on OA with $OX:XA=2:1$, and the point Y lies on OB with $OY:YB=1:2$. The line segments AY and BX intersect at P.



(a) By expressing $\overrightarrow{OP}$ in two ways using the ratio theorem, find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [6]

(b) Find $AP:PY$ and $BP:PX$. [2]

(c) The point Z divides AB internally with $AZ:ZB=1:4$. Show that O, P, Z are collinear and find $OP:PZ$. [2]

END OF HOMEWORK